

SuperFoodsMax Revenue Growth Strategy

DAT107-2026-MARA

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BUSINESS QUESTION

SuperFoodsMax decision-makers ask the data team to provide insights that can support a strategy to lift sales revenue by 5% over the next two years.

Key personnel suggest focusing on retaining loyal customers and converting new customers into loyal customers.

They reveal that:

- A failure to lift the average spend of loyal customers is stagnating revenue growth
- First-time customers are not converting into regular (loyal) customers.

SUB BUSINESS QUESTIONS

1. Customer Spend (Loyal Customers)

- ❑ How has the **average spent of loyal customers** changed from 2019–2022?
- ❑ Which **product categories (commodity/department)** drive the highest spend among loyal customers?
- ❑ What is the **revenue contribution of loyal customers** compared to others?

2. Customer Retention (First-Time → Loyal)

- ❑ What percentage of **first-time customers** convert into loyal customers?
- ❑ How long does it take for a customer to become **loyal**?
- ❑ What purchasing patterns influence **customer retention**?

3. Customer Behaviour Analysis

- ❑ What are the **top purchased commodities** by each loyalty segment?
- ❑ Do **promiscuous (non-loyal) customers** spend more per transaction?
- ❑ How does **customer spending behaviour** vary across age groups and household types?

4. Revenue Growth Drivers

- ❑ Which **products, brands, or departments** contribute most to revenue?
- ❑ What are the **monthly/yearly revenue trends (2019–2022)**?
- ❑ Which customer segments offer the **highest growth potential**?

ACQUIRE DATA

Dataset Summary

- Includes customer, product, and transaction data
- Covers loyalty, age band, household type, department, brand, commodity, store, transaction date, and price
- Supports analysis of revenue growth, customer retention, and product purchasing patterns



Data Sources



SuperFoodsMax Internal Data (Simulated Retail Dataset)



Provided as part of the RMIT Online Data Analytics Program



Represents a retail grocery dataset for **2019–2022**



The analysis uses **dataset_2019_2022.csv**

CLEANING & FORMATTING DATA

Objective

Ensure data is accurate, consistent, and ready for analysis by identifying and fixing data quality issues.



CLEANING & FORMATTING DATA

Step	Data Issues Identified	Data Cleaning Approach
1	Data type inconsistency in transaction_date	Converted transaction date to proper datetime format
2	Inconsistent date format	Standardised transaction_date into a consistent datetime format for time-based analysis
3	Missing / null values	Checked using isnull() and confirmed no missing values in the dataset
4	Duplicate records	Identified duplicate rows using duplicated() and removed them using drop_duplicates()
5	Categorical value inconsistency in loyalty	Standardised loyalty values for consistency and business clarity using replace()
6	Data type optimisation for categorical columns	Converted categorical columns to category type using astype("category") to improve performance
7	Data validation checks	Verified cleaned data using shape, unique(), isnull().sum(), and describe()

DATA ANALYTICS USING PYTHON

- Refer to Section 1 of the *FinalSubmissionCode.ipynb* for dataset loading and data cleaning steps.
- Refer to Section 2 of the *FinalSubmissionCode.ipynb* for Data Visualisation steps.

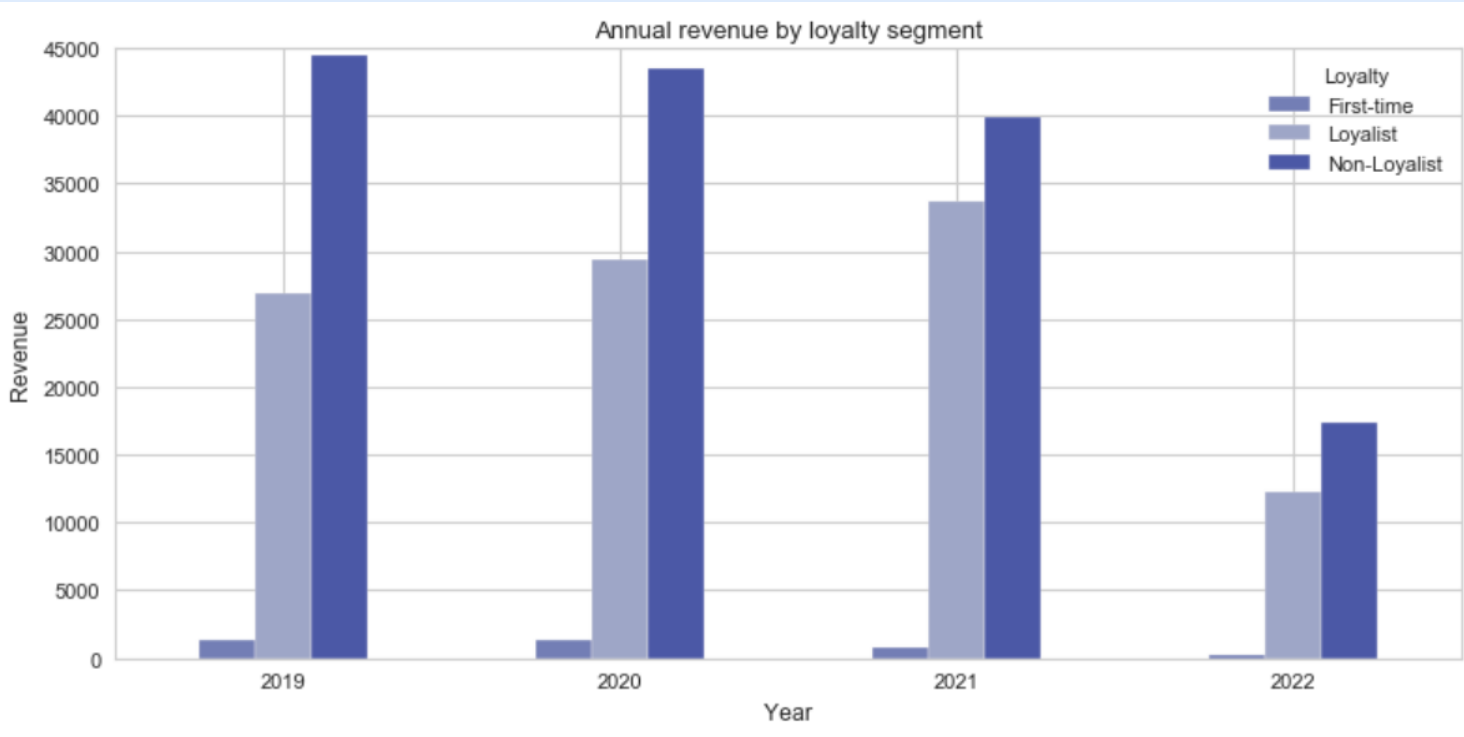
VISUALISATION

Objective

To present data through effective visualisations that highlight trends, patterns, and opportunities for revenue growth.



CUSTOMER LOYALTY IMPACT ON REVENUE GROWTH



Note: 2022 data represents a partial year and is excluded from trend analysis

☐ Key Insights

- Non-Loyalist customers contribute the highest revenue across all years
- Loyalist customers show consistent growth from 2019 to 2021
- First-time customers contribute minimal revenue

☐ Business Interpretation

- Strong reliance on non-loyal customers may impact long-term stability
- Growth opportunity lies in increasing loyal customer spending
- Converting non-loyal customers to loyal customers can drive revenue growth

☐ Recommendation (Linked to 5% Goal)

- Increase average spend of loyal customers
- Implement strategies to convert high-value non-loyal customers
- Improve customer retention and repeat purchases

AVERAGE SPEND PER CUSTOMER BY LOYALTY SEGMENT

❑ Key Insights

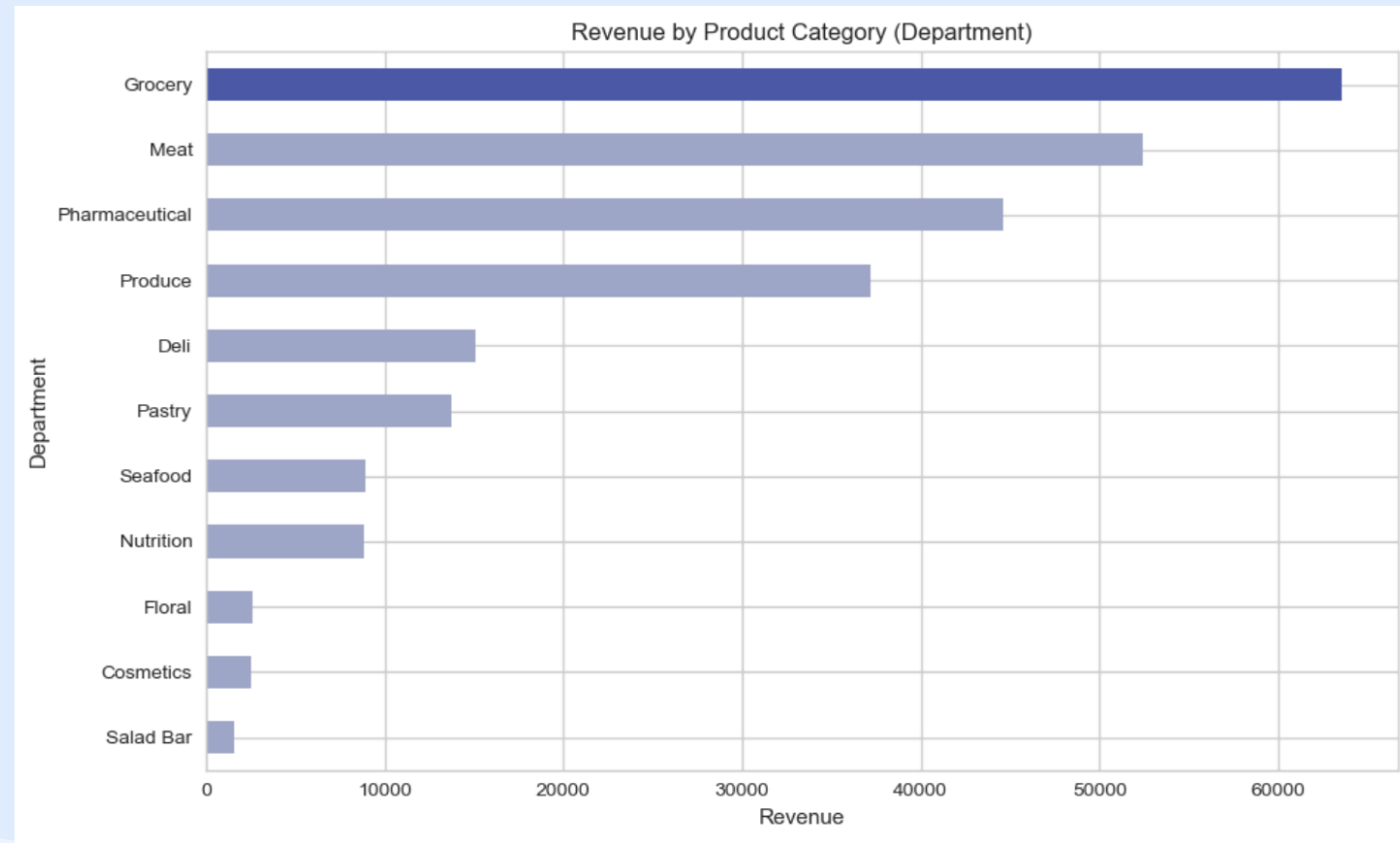
- Grocery category contributes the highest revenue
- Meat and Pharmaceutical categories are strong performers
- Produce shows moderate revenue contribution
- Several categories contribute minimal revenue

❑ Business Interpretation

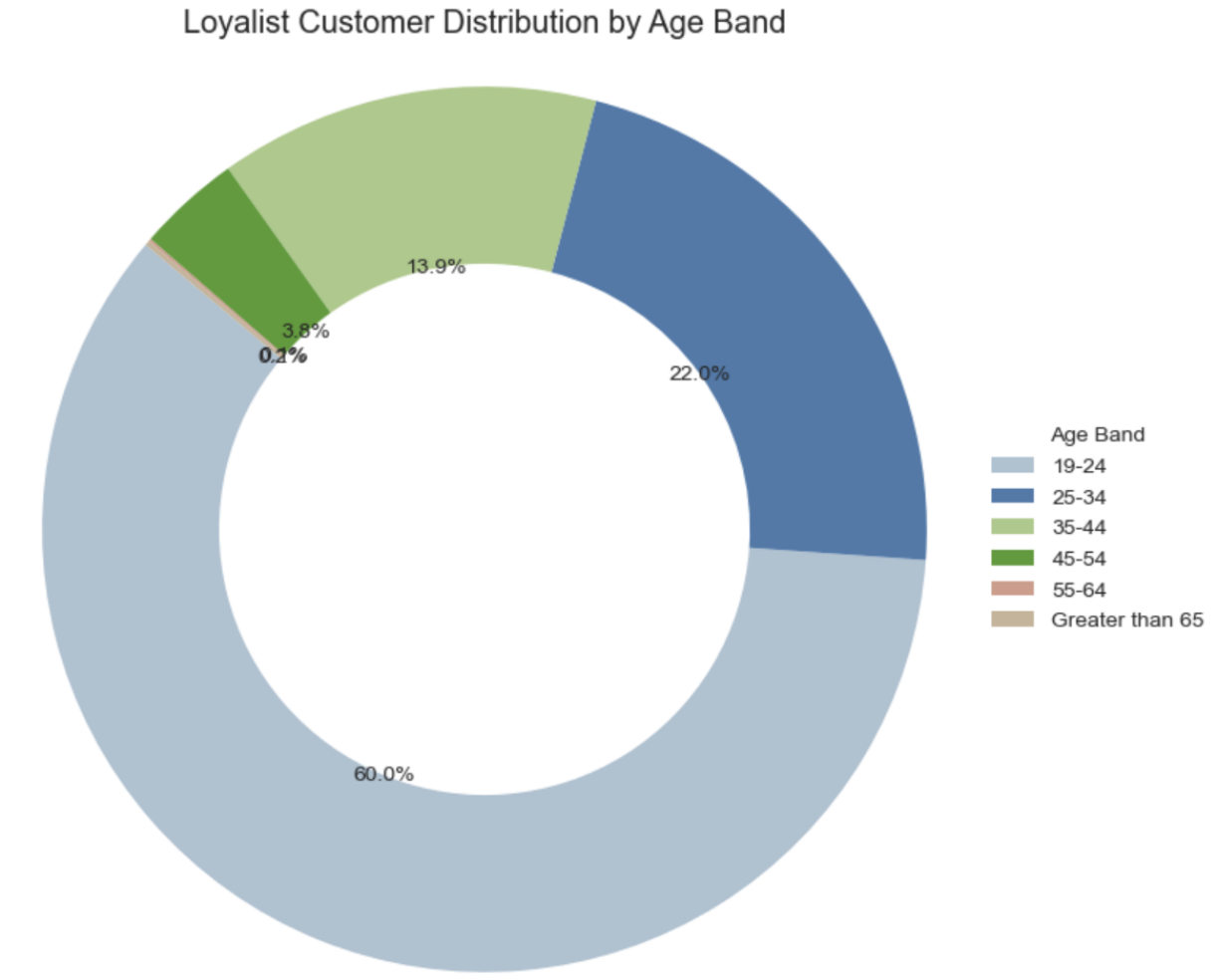
- Revenue is highly concentrated in a few categories
- Growth potential is highest in top-performing categories
- Mid-level categories like Produce offer scalable opportunities
- Low-performing categories may have limited impact on growth

❑ Recommendation

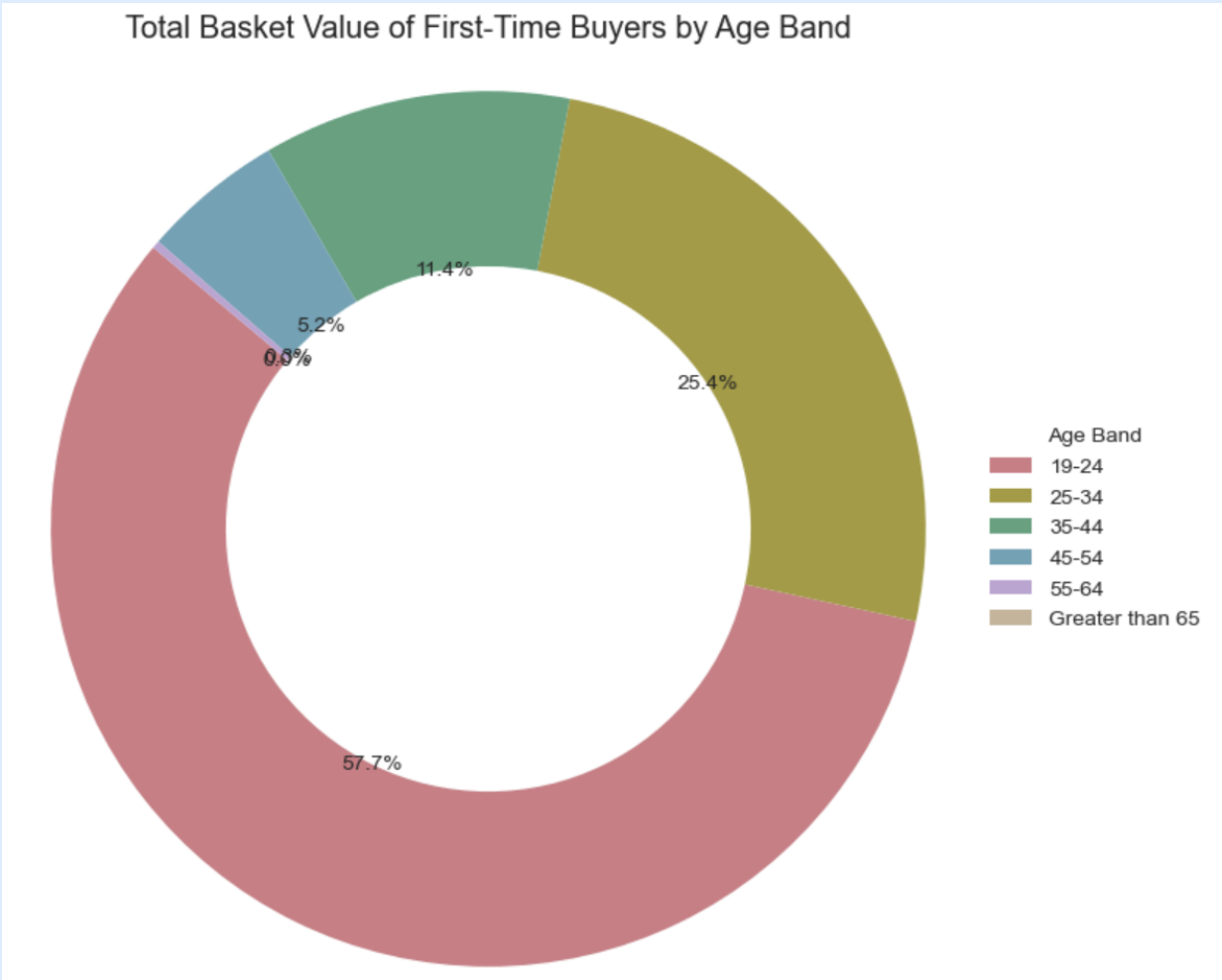
- Focus promotions on Grocery, Meat, and Pharmaceutical
- Use cross-selling strategies to increase basket size
- Improve visibility and offers for Produce category
- Re-evaluate or bundle low-performing categories



LOYALIST CUSTOMER DISTRIBUTION BY AGE BAND



TOTAL BASKET VALUE OF FIRST-TIME BUYERS BY AGE BAND



CUSTOMER AGE PROFILE COMPARISON: LOYALISTS VS FIRST-TIME BUYERS

❑ Key Insights

- Both Loyalists (60%) and First-Time Buyers (57.7%) are heavily concentrated in the 19–24 age group
- The 25–34 segment is the second largest contributor in both groups
- Contribution significantly declines after age 35 in both segments
- Older age groups (45+) contribute minimal share in both acquisition and retention

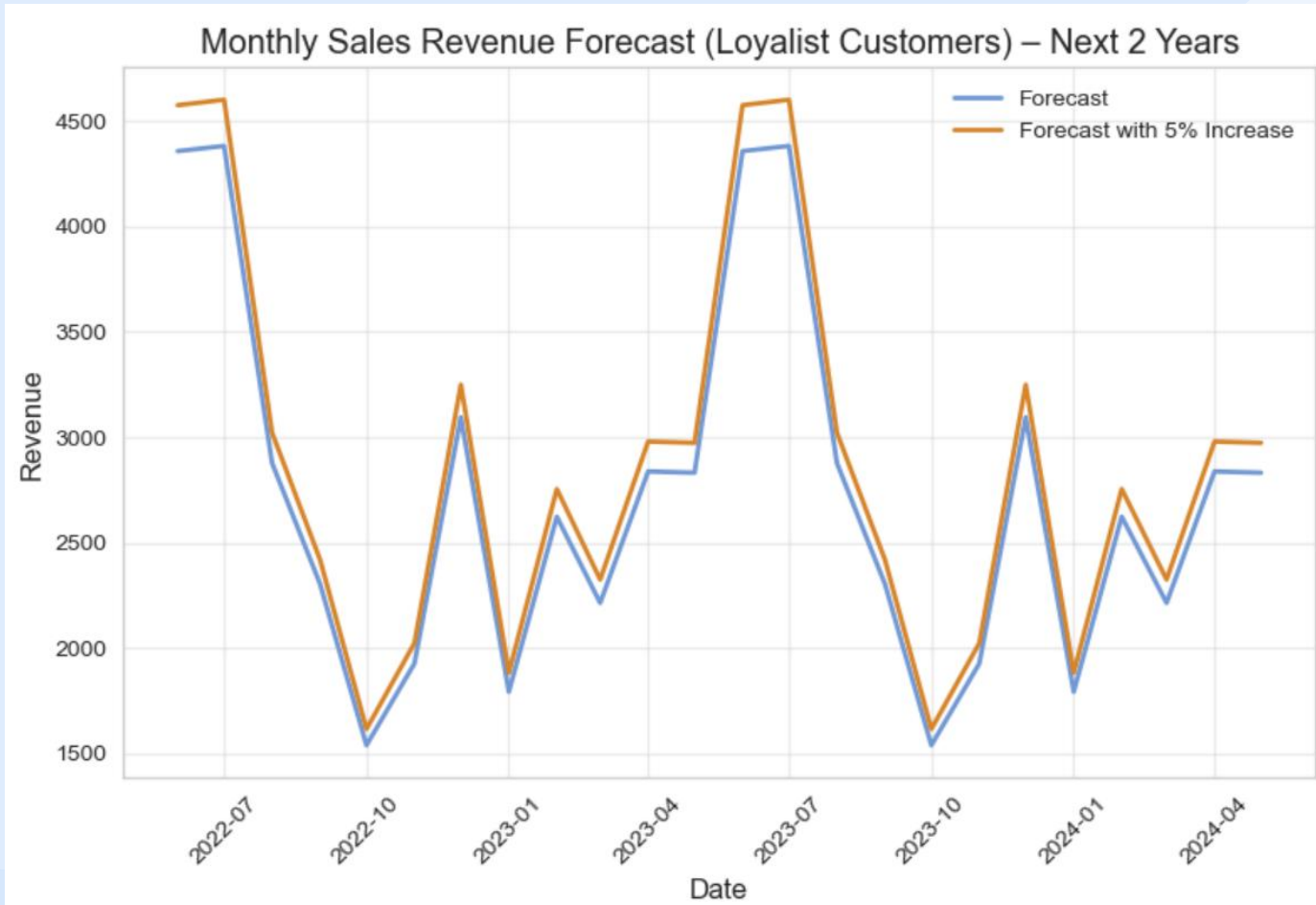
❑ Business Interpretation

- The business is highly dependent on younger customers (19–34) for both revenue and loyalty
- Current strategy may be too narrowly focused on younger demographics, creating long-term risk

❑ Recommendation

- Strengthen retention strategies for 19–34 age group
- Introduce loyalty rewards, personalised offers, and repeat purchase incentives
- Improve Conversion (First-Time → Loyalist)
- Target young first-time buyers with:
 - Follow-up promotions
 - Membership/loyalty programs
- Focus on increasing repeat purchases
- Develop targeted campaigns for mid-age and older customers
- Short-term: Drive revenue via young loyal customers
- Long-term: Build stability by diversifying customer age base

MONTHLY SALES REVENUE FORECAST (LOYALIST CUSTOMERS) – NEXT 2 YEARS



□ Key Insights

- Monthly revenue shows a recurring seasonal pattern with peaks and dips
- Forecast indicates stable but limited natural growth without intervention
- A 5% increase scenario consistently lifts revenue across all months
- The gap between baseline and 5% scenario becomes more significant over time

□ Business Interpretation

- Loyalist revenue alone is not growing fast enough to meet business targets
- Revenue fluctuations suggest seasonal buying behaviour
- Even a small improvement (5%) can create meaningful long-term impact
- Growth is achievable but requires active strategy, not passive trend

FINAL RECOMMENDATIONS & STRATEGIC DIRECTION FOR 5% REVENUE GROWTH

❑ Overall Conclusion

- The current revenue trend shows strong reliance on Non-Loyalist customers
- Loyalist customers provide stable and scalable growth potential
- First-time buyers represent a missed opportunity due to low conversion
- Revenue is highly concentrated in a few product categories and younger age groups

❑ Key Strategic Priorities

1. Grow Loyal Customer Value

- Increase average basket size through bundles and targeted offers
- Focus on repeat purchase behaviour

2. Improve First-Time Buyer Conversion

- Launch post-purchase campaigns (14–30 days)
- Introduce loyalty programs & incentives
- Convert high-potential customers into loyalists

3. Maximise High-Performing Categories

- Focus on Grocery, Meat, and Pharmaceutical
- Use cross-selling and upselling strategies
- Expand mid-tier categories like Produce

4. Target High-Value Customer Segments

- Prioritise 19–34 age group (largest contributors)
- Develop strategies to engage older age groups

FINAL BUSINESS RECOMMENDATION

Achieving 5% revenue growth is realistic. But requires targeted, data-driven action, not broad discounting.

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THANK YOU